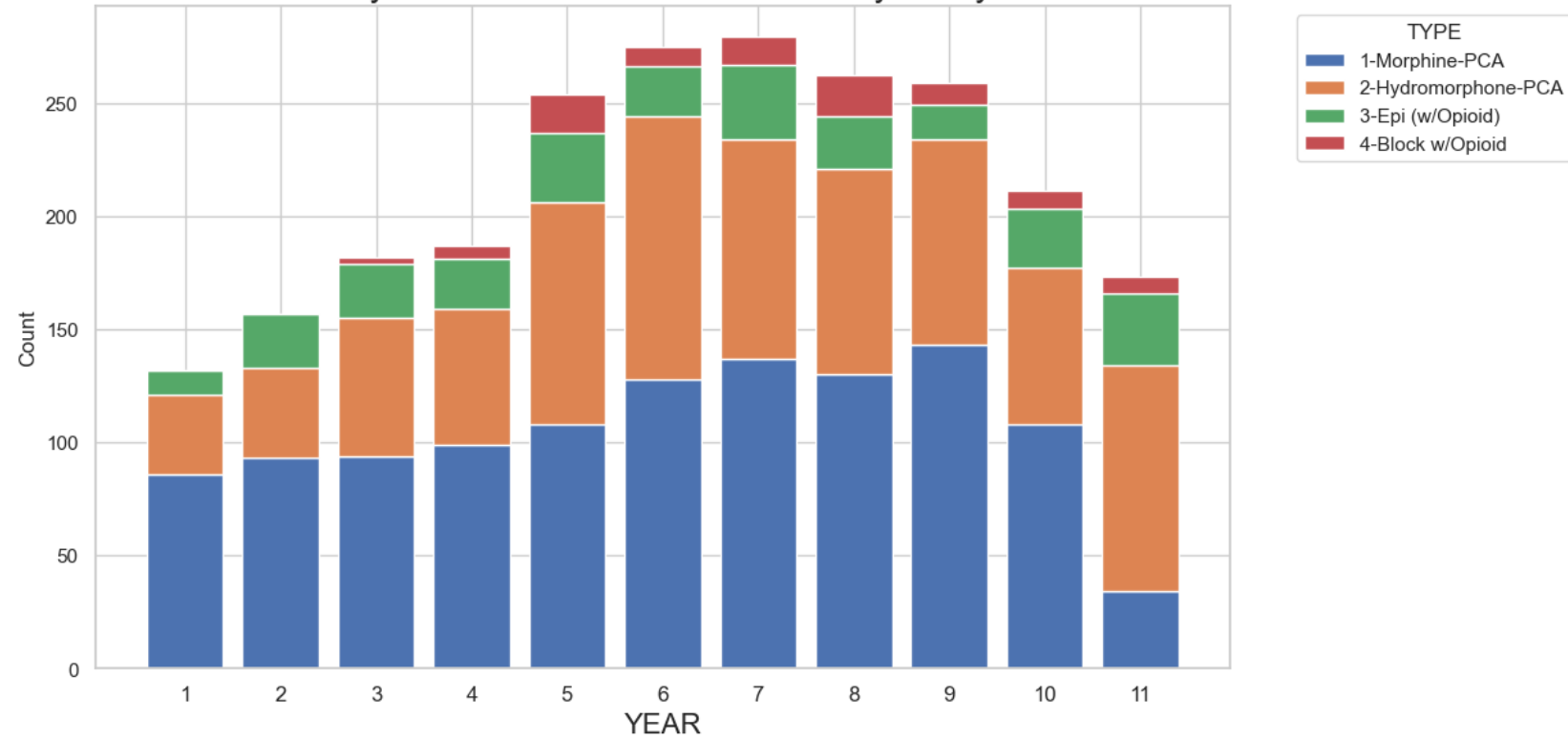
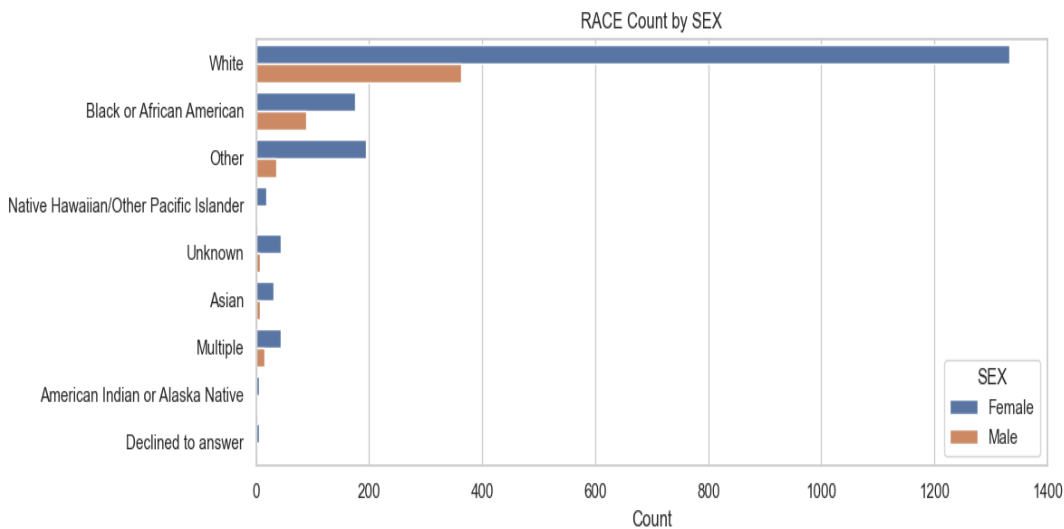
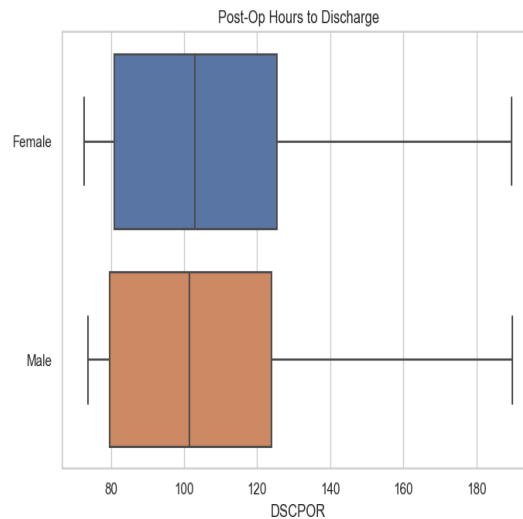
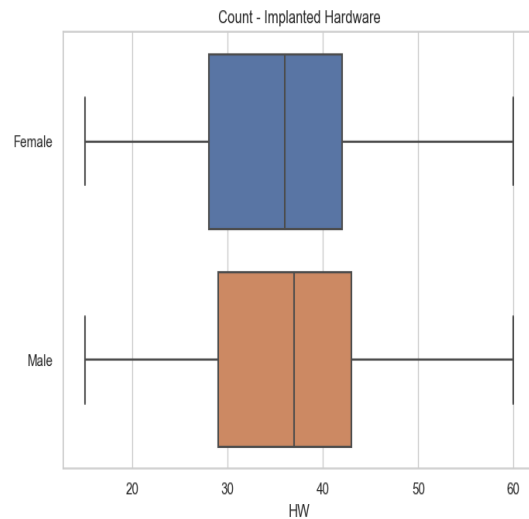
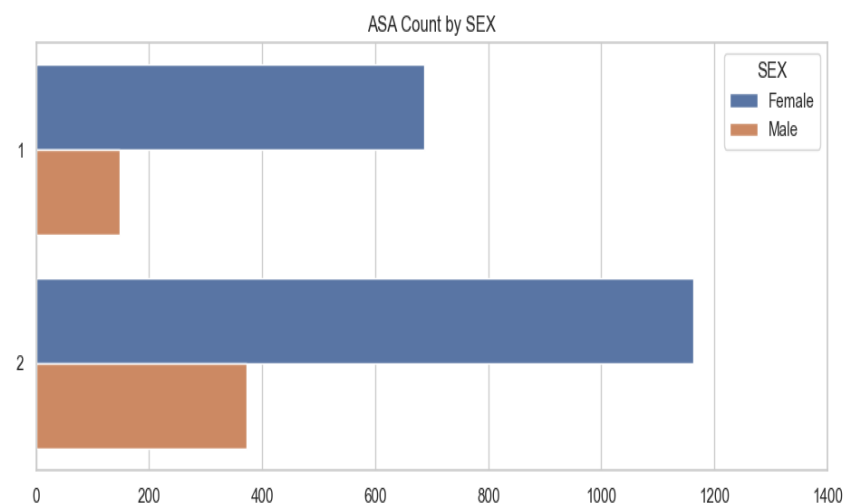
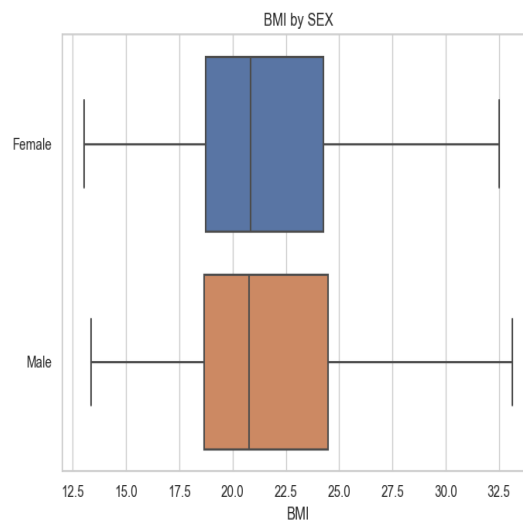
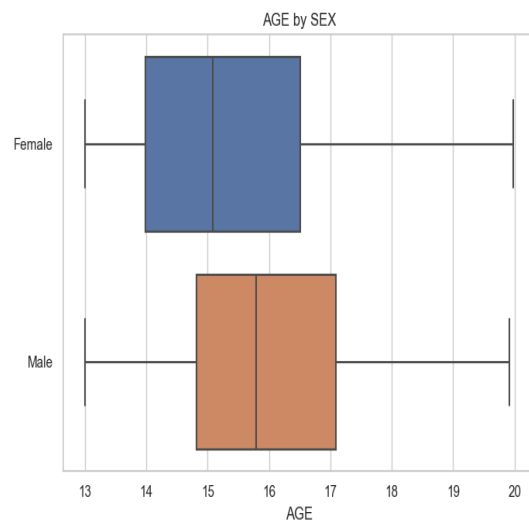
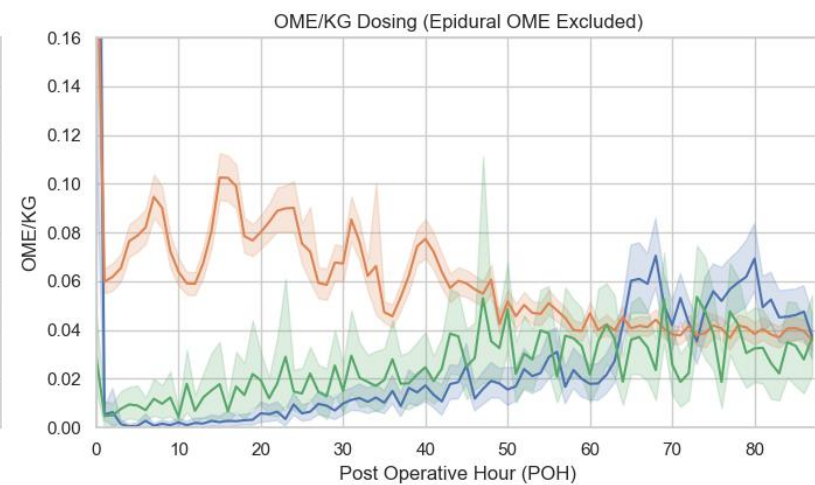
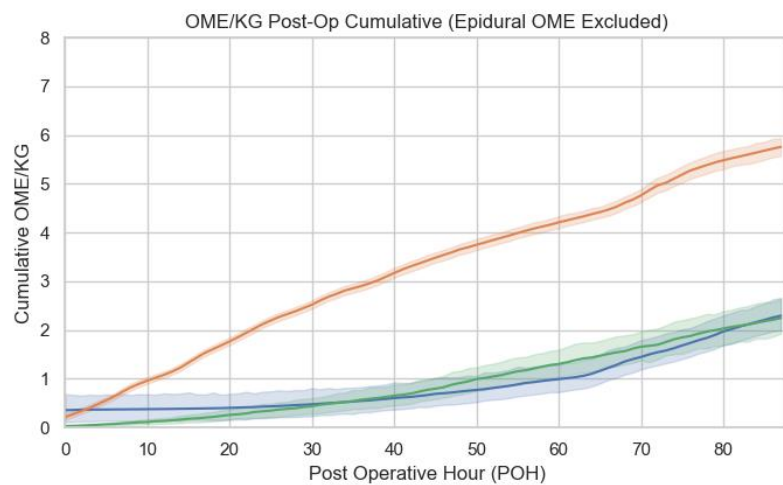
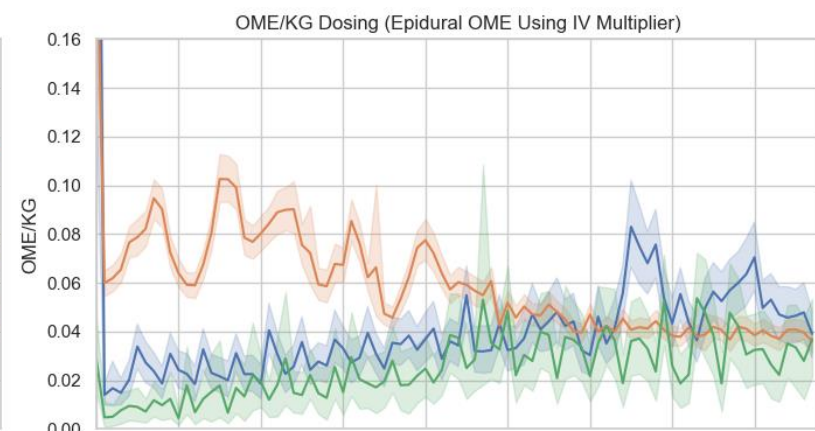
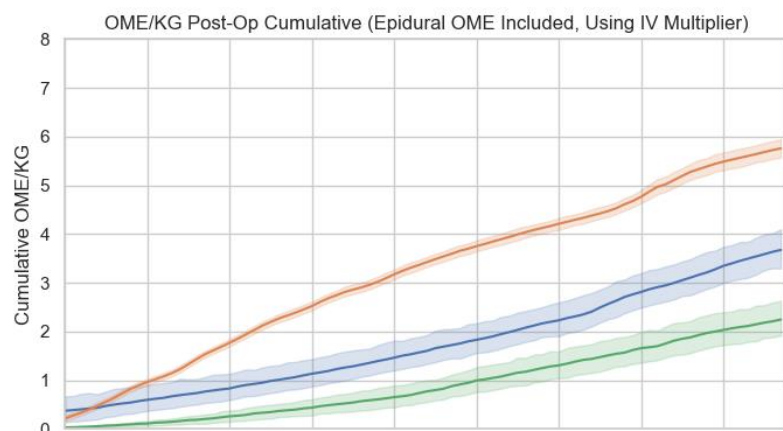
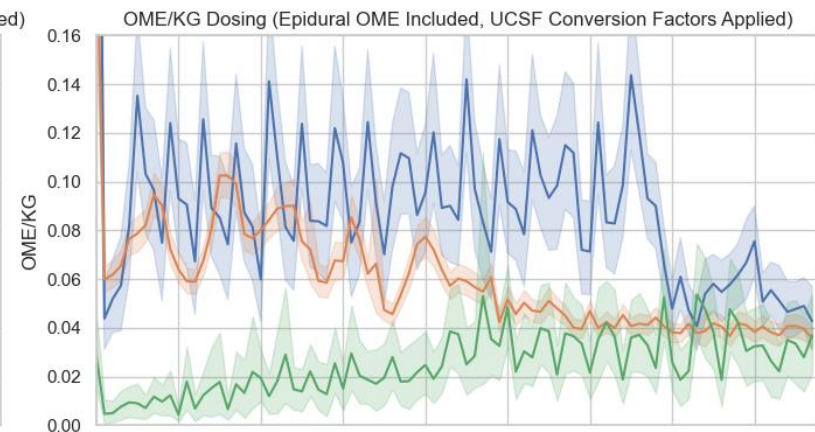
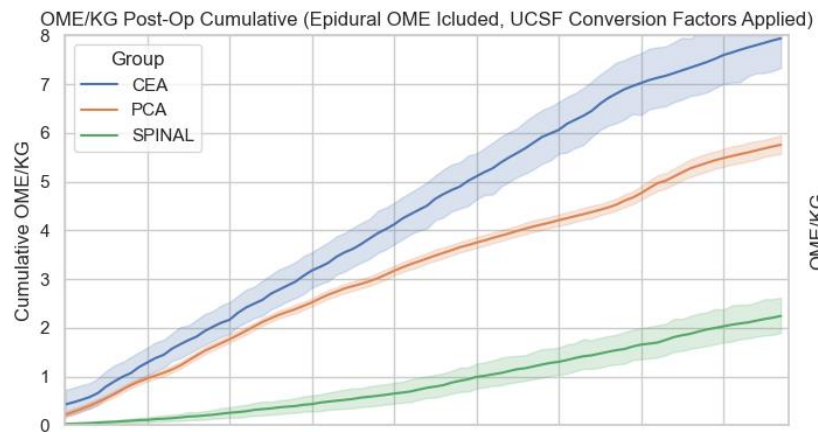


Counts by Patient TYPE and FACILITY by Study YEAR

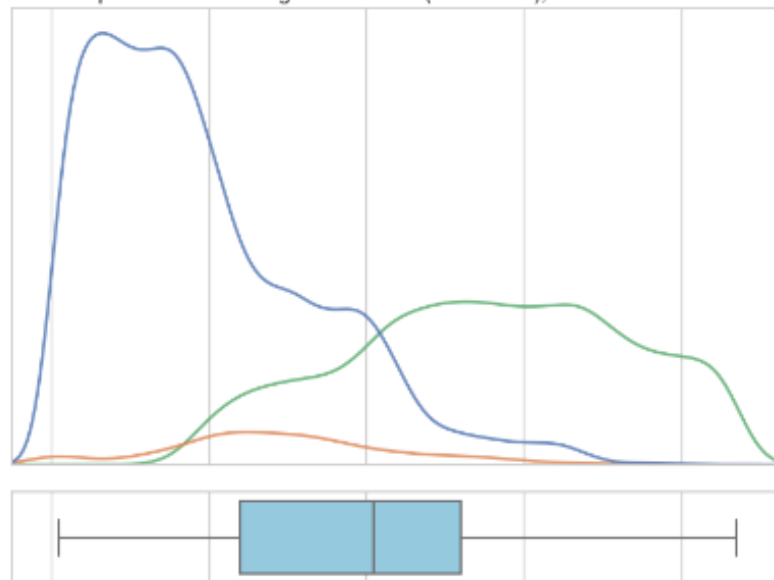


	1	2	3	4	5	6	7	8	9	10	11
CHICAGO	19	20	31	12	22	23	21	28	26	19	22
GREENVILLE	31	39	47	33	34	17	11	11	9	14	9
HONOLULU	0	0	0	2	10	4	7	2	9	7	7
HOUSTON	1	11	9	6	13	2	7	16	15	7	0
LEXINGTON	8	6	13	19	16	16	2	0	0	0	0
NORTHERN CALIFO	0	0	3	4	12	14	15	16	1	1	0
PHILADELPHIA	0	0	0	8	28	43	54	47	52	42	49
PORTLAND	15	11	6	19	15	20	29	26	36	19	18
SHREVEPORT	24	10	22	33	18	30	28	28	43	46	18
SPOKANE	11	24	24	22	31	22	33	23	15	26	32
SPRINGFIELD	10	14	15	8	2	19	21	11	18	3	0
ST. LOUIS	6	10	9	6	24	43	29	28	24	27	18
TAMPA	7	12	3	15	29	22	22	26	11	0	0

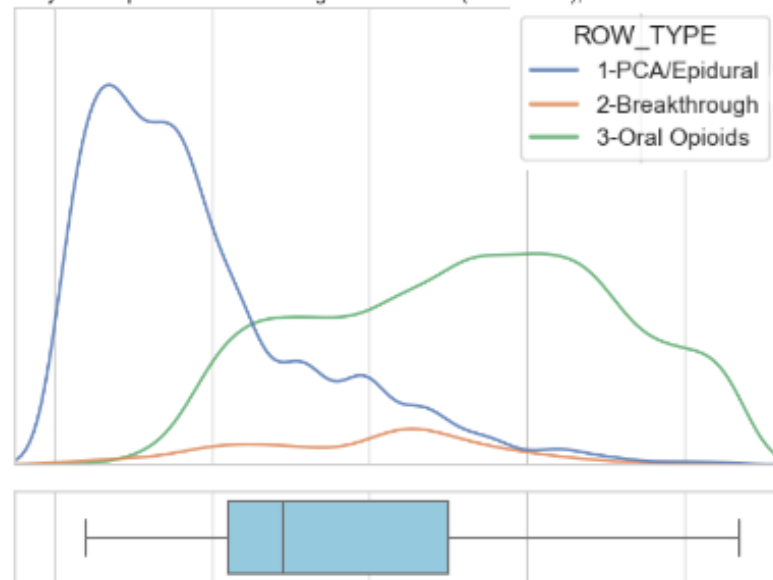




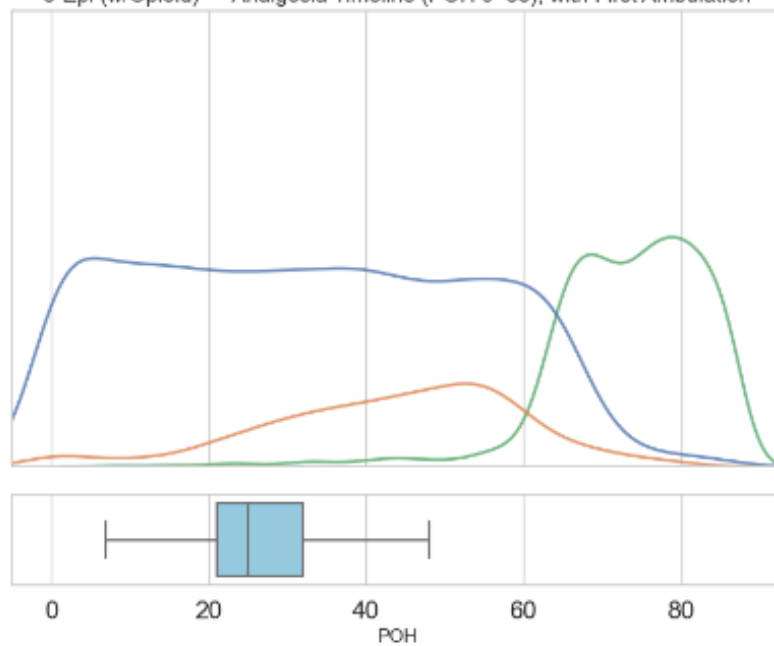
1-Morphine-PCA — Analgesia Timeline (POH 0–88), with First Ambulation



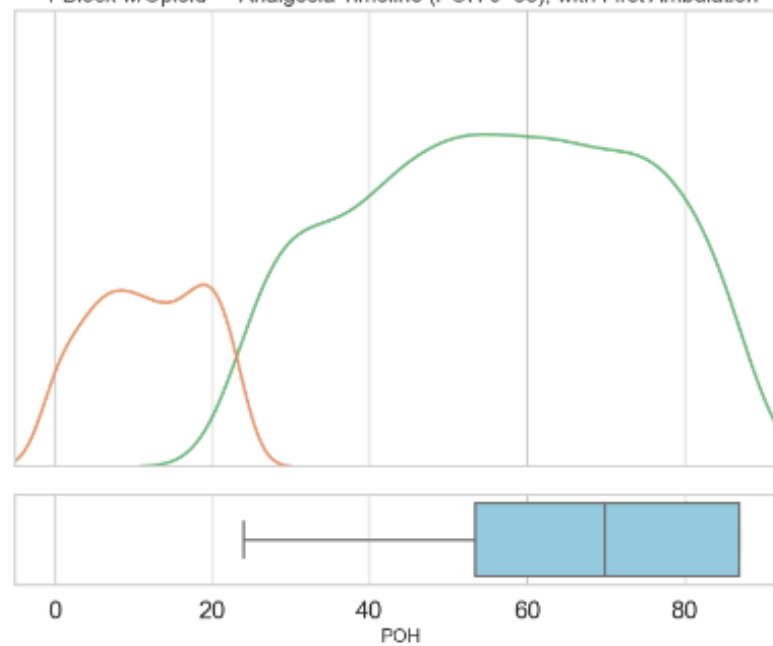
2-Hydromorphone-PCA — Analgesia Timeline (POH 0–88), with First Ambulation

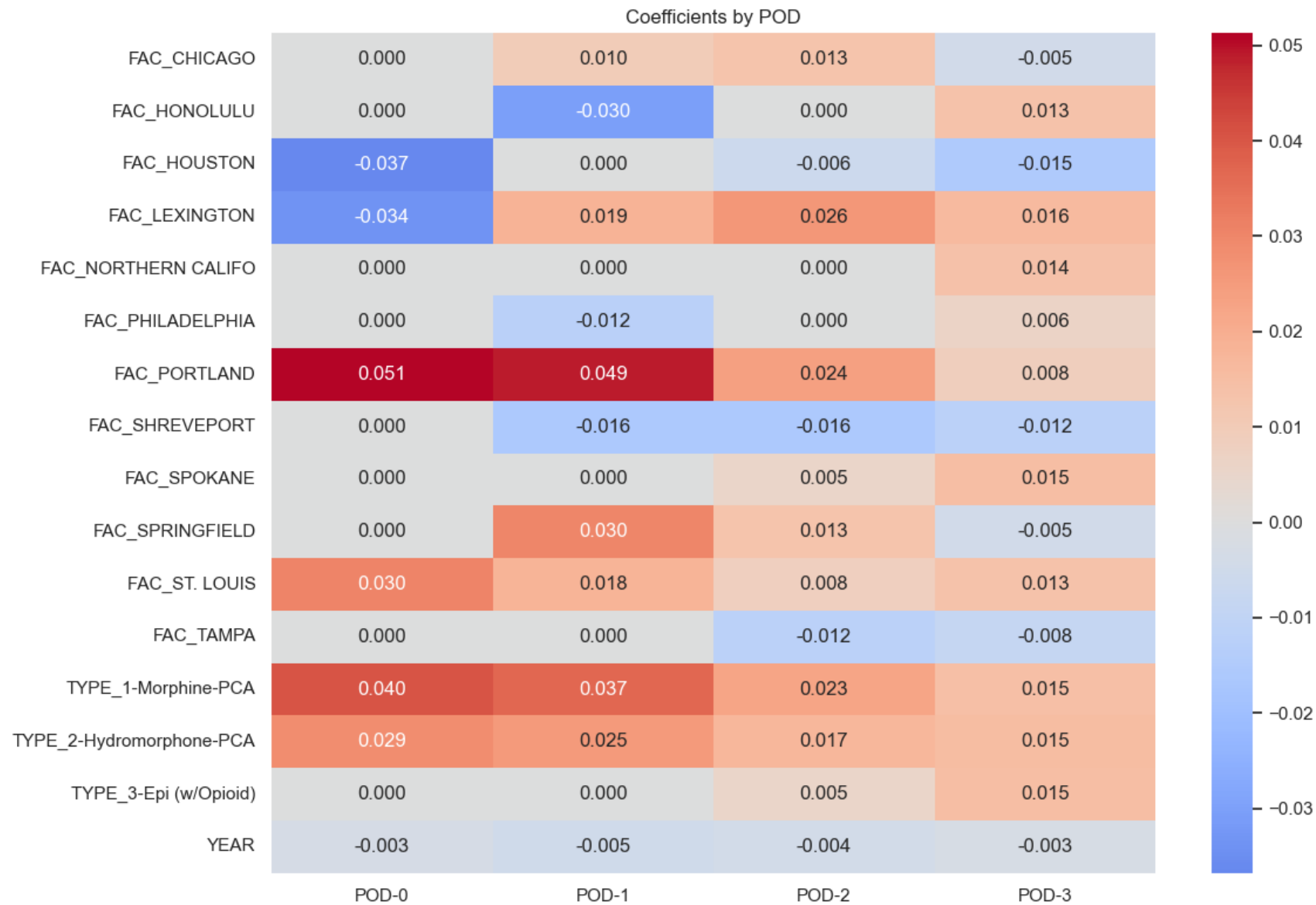


3-Epi (w/Opioid) — Analgesia Timeline (POH 0–88), with First Ambulation

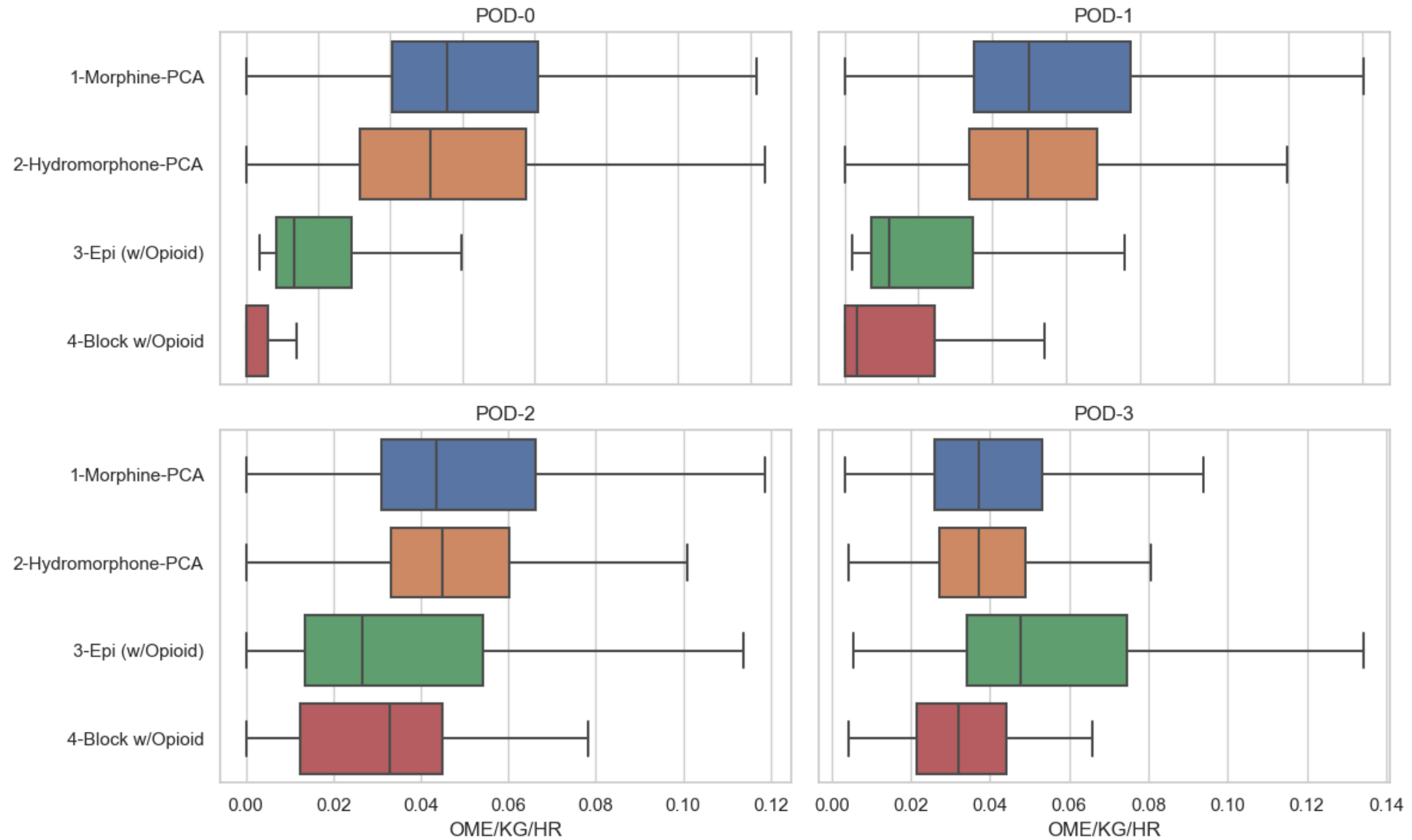


4-Block w/Opioid — Analgesia Timeline (POH 0–88), with First Ambulation

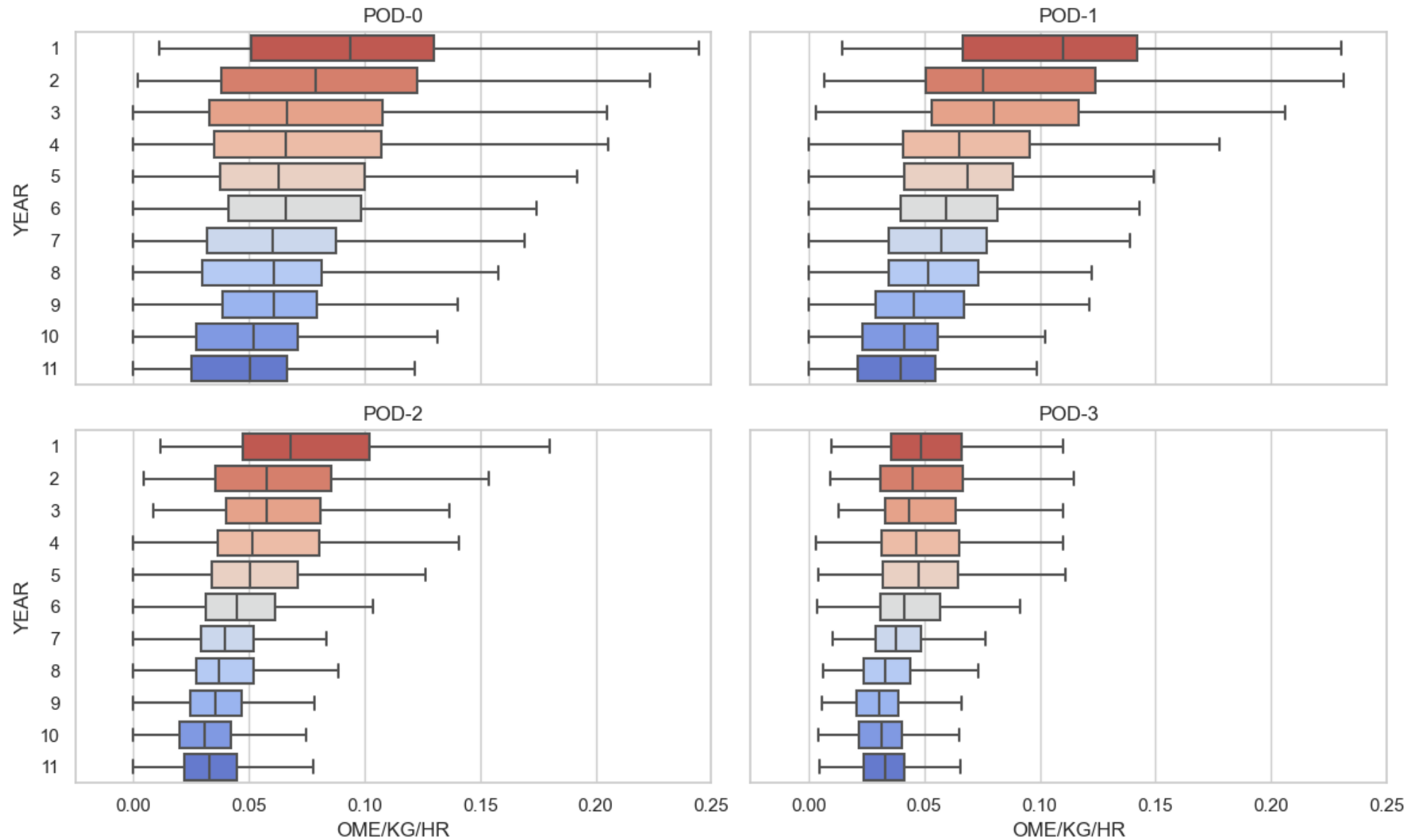




Distribution of OME/KG/HR by TYPE Across PODs



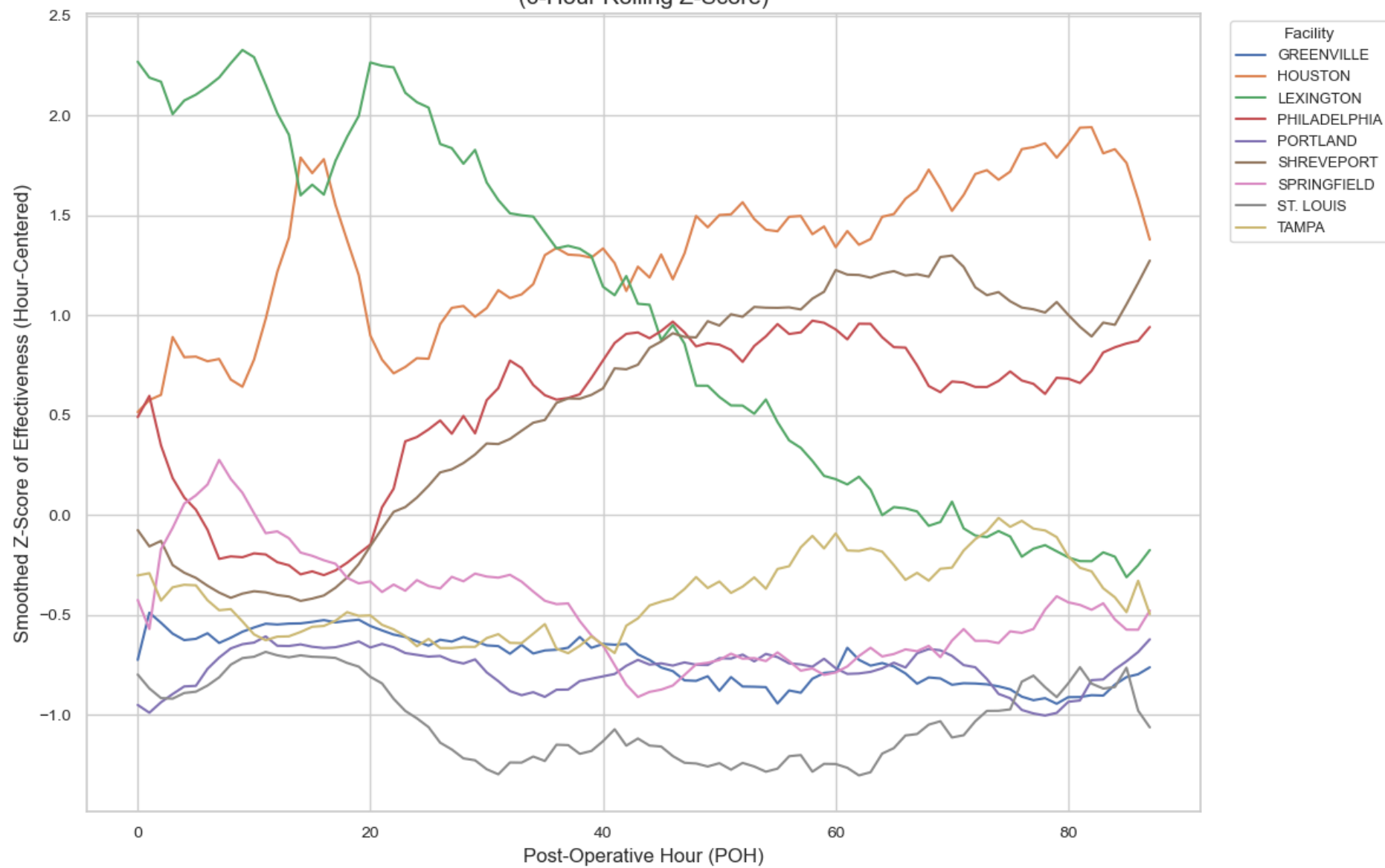
Distribution of OME/KG/HR by YEAR and POD



- Mann–Whitney U test (median difference)
 - Kolmogorov–Smirnov test (distributional difference)
 - Cliff's Delta (effect size)
-
- $\geq 20\%$ reduction → Clinically significant
 - 10–19.99% reduction → Possibly significant
 - $< 10\%$ reduction or increase → Not clinically significant

	POD	Compared TYPE	Mann–Whitney p	K–S p	Cliff's Delta	Median Morphine	Median Compared	% Difference	Clinical Interpretation
0	POD-0	2-Hydromorphone-PCA	0.0000	0.0000	0.111000	0.069700	0.063900	-8.30%	Not clinically significant
1	POD-0	3-Epi (w/Opioid)	0.0000	0.0000	0.709000	0.069700	0.016800	-75.83%	Clinically significant
2	POD-0	4-Block w/Opioid	0.0000	0.0000	0.861000	0.069700	0.000000	-100.00%	Clinically significant
3	POD-1	2-Hydromorphone-PCA	0.0365	0.0004	0.054000	0.062500	0.061900	-0.95%	Not clinically significant
4	POD-1	3-Epi (w/Opioid)	0.0000	0.0000	0.678000	0.062500	0.015100	-75.86%	Clinically significant
5	POD-1	4-Block w/Opioid	0.0000	0.0000	0.800000	0.062500	0.004100	-93.52%	Clinically significant
6	POD-2	2-Hydromorphone-PCA	0.9576	0.0023	0.001000	0.043400	0.045000	+3.68%	Not clinically significant
7	POD-2	3-Epi (w/Opioid)	0.0000	0.0000	0.327000	0.043400	0.026600	-38.65%	Clinically significant
8	POD-2	4-Block w/Opioid	0.0000	0.0000	0.374000	0.043400	0.033000	-24.01%	Clinically significant
9	POD-3	2-Hydromorphone-PCA	0.9260	0.0016	-0.002000	0.037100	0.037200	+0.09%	Not clinically significant
10	POD-3	3-Epi (w/Opioid)	0.0000	0.0000	-0.299000	0.037100	0.047800	+28.70%	Clinically significant
11	POD-3	4-Block w/Opioid	0.0009	0.0093	0.212000	0.037100	0.032100	-13.59%	Possibly significant

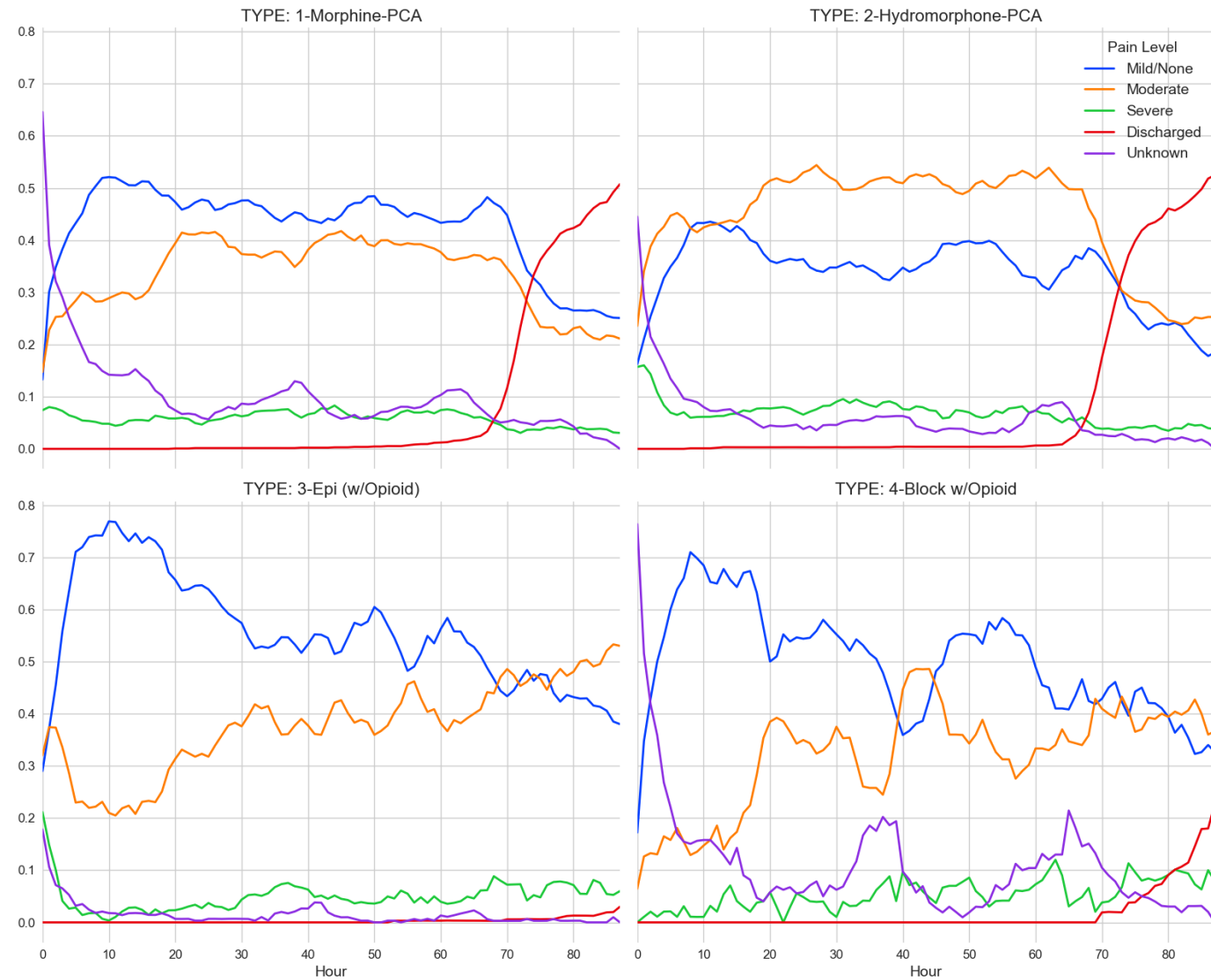
Smoothed Facility-Level Variation in Morphine-PCA Effectiveness
(6-Hour Rolling Z-Score)



OME/KG/HR vs SHREVEPORT (n = 300) by POD and Facility (Morphine-PCA Only)				
GREENVILLE	+28.4% (n = 85) Significant	+103.3% (n = 85) Significant	+121.9% (n = 85) Significant	+106.0% (n = 82) Significant
HOUSTON	-69.2% (n = 81) Significant	+40.9% (n = 81) Significant	+13.2% (n = 80) Possibly Significant	-14.2% (n = 79) Invalid (p-val > 0.05)
LEXINGTON	-45.5% (n = 76) Significant	+81.8% (n = 76) Significant	+138.6% (n = 76) Significant	+136.8% (n = 75) Significant
PHILADELPHIA	-17.7% (n = 279) Possibly Significant	-2.1% (n = 279) Invalid (p-val > 0.05)	+35.9% (n = 278) Significant	+52.5% (n = 276) Significant
PORTLAND	+57.3% (n = 172) Significant	+101.4% (n = 172) Significant	+105.3% (n = 170) Significant	+86.1% (n = 168) Significant
SPRINGFIELD	-9.3% (n = 81) Invalid (p-val > 0.05)	+110.2% (n = 81) Significant	+113.0% (n = 81) Significant	+55.0% (n = 81) Significant
ST. LOUIS	+90.7% (n = 53) Significant	+107.7% (n = 53) Significant	+89.1% (n = 53) Significant	+107.9% (n = 53) Significant
TAMPA	+7.1% (n = 33) Invalid (p-val > 0.05)	+17.4% (n = 33) Invalid (p-val > 0.05)	+5.5% (n = 33) Insignificant	+19.4% (n = 32) Possibly Significant
	POD-0	POD-1	POD-2	POD-3

% Change in Scaled Pain Grade vs SHREVEPORT (Morphine-PCA Only, with 95% CI)				
GREENVILLE	+16.4% CI: ±0.03	+16.6% CI: ±0.03	+16.9% CI: ±0.03	+17.0% CI: ±0.03
HOUSTON	-10.9% CI: ±0.03	-8.5% CI: ±0.02	-5.7% CI: ±0.02	-5.8% CI: ±0.03
LEXINGTON	+3.8% CI: ±0.03	+11.1% CI: ±0.03	+12.0% CI: ±0.03	+10.2% CI: ±0.03
PHILADELPHIA	+6.4% CI: ±0.03	+6.5% CI: ±0.02	+7.5% CI: ±0.02	+4.1% CI: ±0.02
PORTLAND	+12.1% CI: ±0.02	+3.2% CI: ±0.02	+6.7% CI: ±0.02	+10.1% CI: ±0.02
SPRINGFIELD	+25.1% CI: ±0.04	+27.1% CI: ±0.03	+24.3% CI: ±0.03	+20.3% CI: ±0.03
ST. LOUIS	+33.7% CI: ±0.04	+17.5% CI: ±0.03	+15.8% CI: ±0.03	+12.0% CI: ±0.04
TAMPA	+32.6% CI: ±0.06	+17.7% CI: ±0.04	+16.0% CI: ±0.04	+14.0% CI: ±0.05
	POD-0	POD-1	POD-2	POD-3

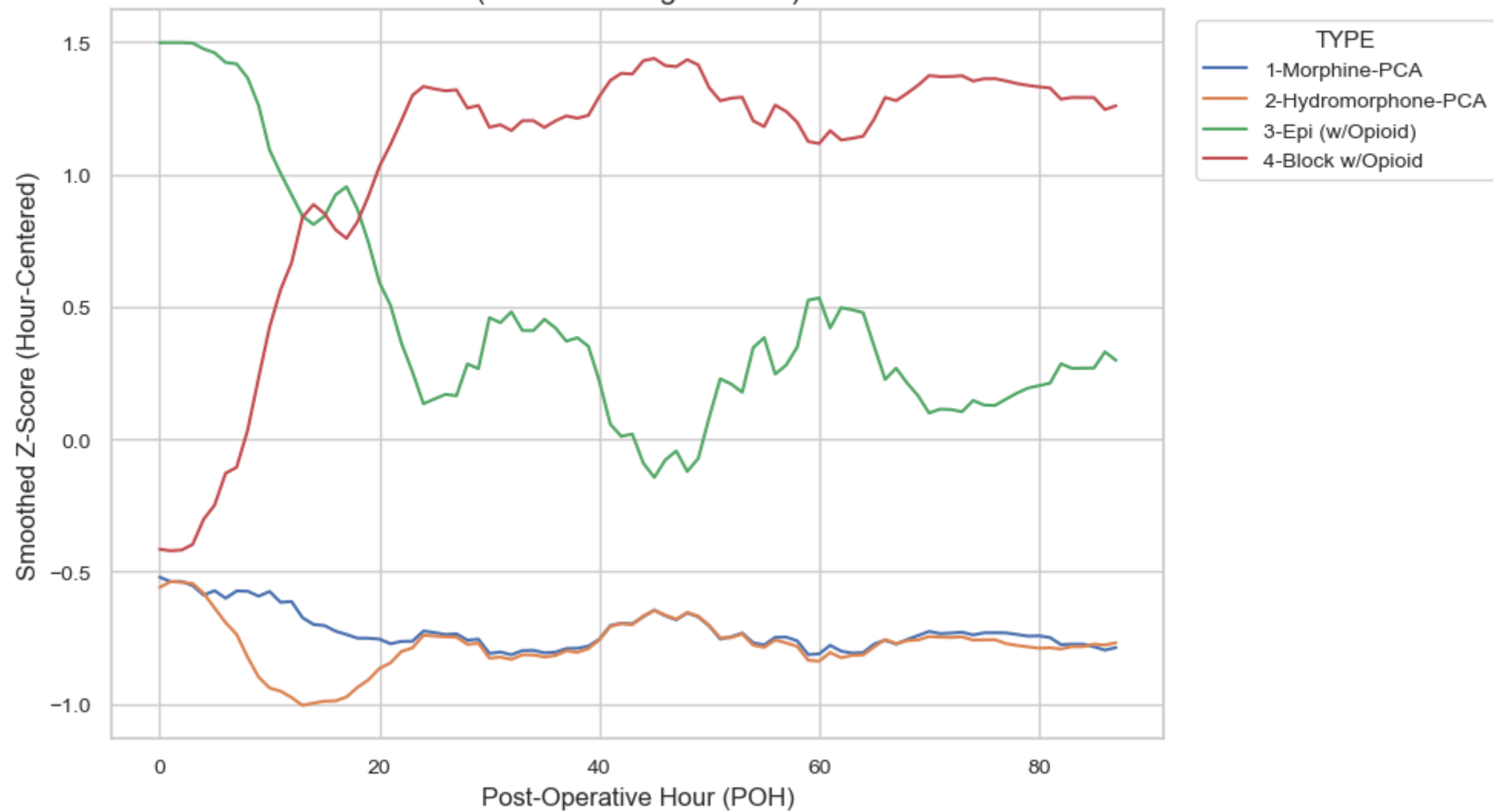
Pain Grade Proportions by Hour, Split by TYPE



% Change in Scaled Pain Grade vs 1-Morphine-PCA
(With 95% CI - Scaled Grade Units)

2-Hydromorphone-PCA	+11.6% CI: ± 0.01	+10.0% CI: ± 0.01	+7.1% CI: ± 0.01	+6.1% CI: ± 0.01
3-Epi (w/Opioid)	-8.8% CI: ± 0.02	-7.9% CI: ± 0.01	-5.6% CI: ± 0.01	+4.1% CI: ± 0.01
4-Block w/Opioid	-16.4% CI: ± 0.03	-7.1% CI: ± 0.02	-2.5% CI: ± 0.03	+4.6% CI: ± 0.03
	POD-0	POD-1	POD-2	POD-3

Smoothed Standardized Effectiveness by Hour
(6-Hour Rolling Z-Score)



Standardized Effectiveness by POD and YEAR (Standard Deviations)

YEAR	1	-1.30	-1.57	-1.83	-1.67
	2	0.21	-0.80	-0.99	-0.91
	3	-0.73	-1.03	-0.96	-1.01
	4	0.65	1.91	-0.14	-0.74
	5	-0.95	0.82	0.41	-0.05
	6	-1.10	-0.70	-0.46	-0.12
	7	0.57	0.20	0.61	0.73
	8	1.13	0.37	0.70	0.94
	9	-0.10	-0.37	0.01	0.28
	10	1.88	0.46	1.35	1.47
	11	-0.26	0.71	1.28	1.08
		POD-0	POD-1	POD-2	POD-3

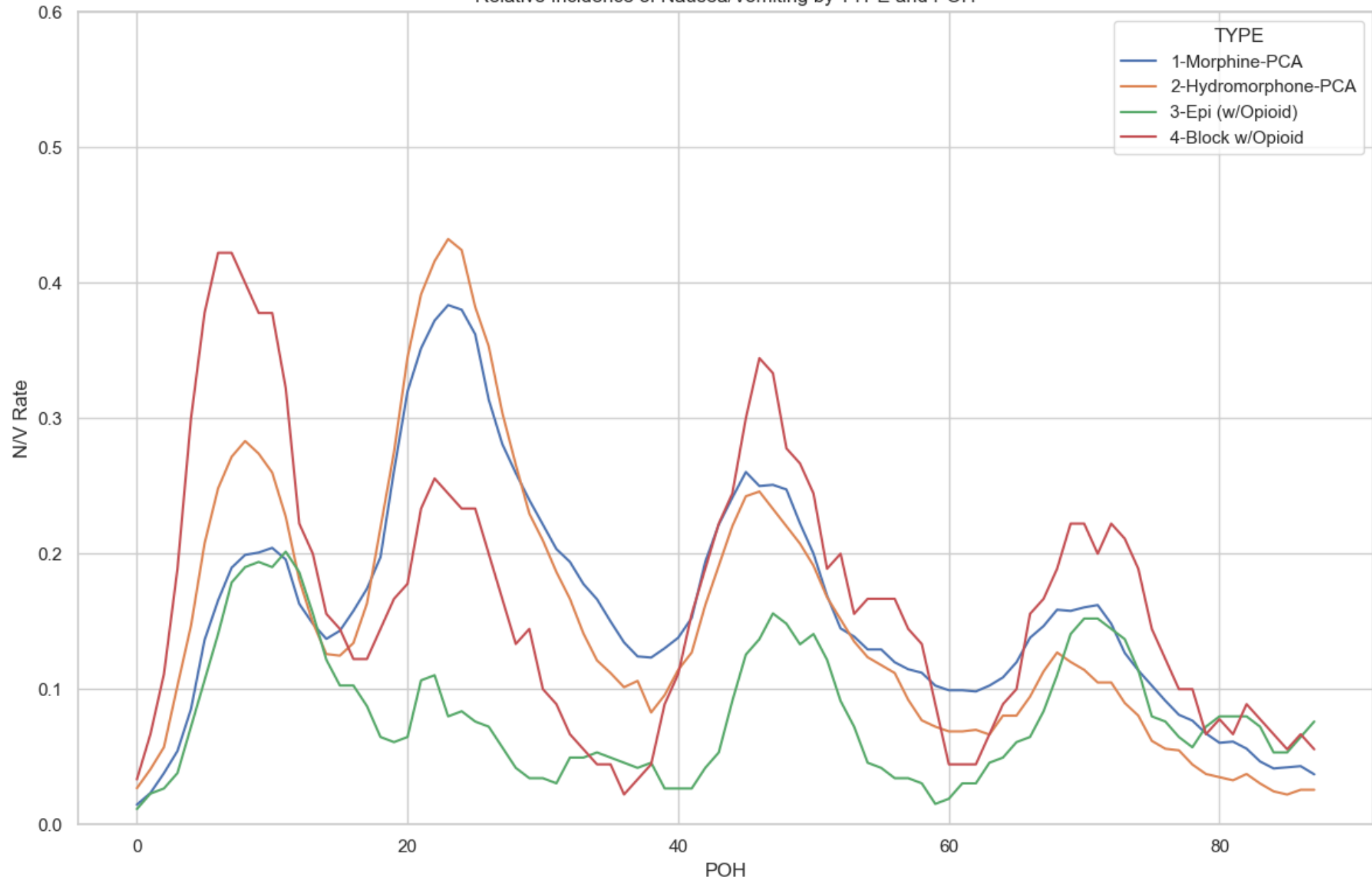
Mann–Whitney U-Test: Complication Rate Differences vs 1-Morphine-PCA by POD and TYPE

Spasm				NV				Constipation				
2-Hydromorphone-PCA	+10.85% p=0.000 Marginal	+18.58% p=0.000 Marginal	+13.00% p=0.000 Marginal	+7.30% p=0.000 Slight	+3.92% p=0.000 Slight	-0.09% p=0.817 Insignificant	-1.92% p=0.000 Slight	-3.13% p=0.000 Slight	-0.00% p=0.911 Insignificant	+4.79% p=0.000 Slight	+6.97% p=0.000 Slight	+4.36% p=0.000 Slight
	-16.15% p=0.000 Marginal	-22.69% p=0.000 * Significant *	-27.54% p=0.000 * Significant *	-8.85% p=0.000 Slight	-0.99% p=0.083 Insignificant	-17.55% p=0.000 Marginal	-9.36% p=0.000 Slight	-0.96% p=0.020 Slight	-0.05% p=0.132 Insignificant	+0.84% p=0.000 Slight	-7.84% p=0.000 Slight	+4.61% p=0.000 Slight
	+6.78% p=0.000 Slight	-3.06% p=0.003 Slight	-8.00% p=0.000 Slight	+8.08% p=0.000 Slight	+12.65% p=0.000 Marginal	-10.47% p=0.000 Marginal	+1.51% p=0.069 Insignificant	+2.95% p=0.000 Slight	-0.05% p=0.378 Insignificant	-0.93% p=0.001 Slight	-10.31% p=0.000 Marginal	-6.01% p=0.000 Slight
3-Epi (w/Opioid)												
4-Block w/Opioid												

Pruritis				POAE				Legend			
2-Hydromorphone-PCA	-1.76% p=0.000 Slight	-0.49% p=0.036 Slight	-0.08% p=0.698 Insignificant	+0.71% p=0.000 Slight	-0.38% p=0.076 Insignificant	+0.18% p=0.007 Slight	+0.09% p=0.018 Slight	+0.04% p=0.050 Slight	p ≥ 0.05	→ Insignificant	
	+0.64% p=0.146 Insignificant	-2.86% p=0.000 Slight	-1.31% p=0.000 Slight	-0.28% p=0.137 Insignificant	-1.96% p=0.000 Slight	-0.24% p=0.005 Slight	+0.03% p=0.569 Insignificant	+0.11% p=0.001 Slight	< 10% & p < 0.05	→ Slight	
	+8.64% p=0.000 Slight	+0.63% p=0.274 Insignificant	+1.22% p=0.011 Slight	+4.39% p=0.000 Slight	+1.58% p=0.003 Slight	-0.11% p=0.461 Insignificant	+0.06% p=0.493 Insignificant	+0.20% p=0.000 Slight	10–19.9% & p < 0.05	→ Marginal	
3-Epi (w/Opioid)									≥ 20% & p < 0.05	→ Significant	
4-Block w/Opioid											
POD-0	POD-1	POD-2	POD-3	POD-0	POD-1	POD-2	POD-3				



Relative Incidence of Nausea/Vomiting by TYPE and POH



Relative Incidence of Constipation by TYPE and POH

